

HSIANG-CHUN (NATHAN) WANG

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EDUCATION (PHOTOS)

National Taiwan University (NTU)

09/2022 – 06/2024

M.S. in Communication Engineering

Taipei, Taiwan

- Spearheaded reinforcement learning research, leading to first-author publications at ICML and NeurIPS with **over 90 citations**.
- Won the TAAI Best Master's Thesis Award (top 5 nationwide) and NTU's Best Master's Thesis Award (top 3 in the department).
- Selected as reviewer for ICLR 2025 and NeurIPS 2025–2026, recognizing ML contributions.

Shanghai Jiao Tong University (SJTU)

09/2018 – 08/2022

B.S. in Information Engineering

Shanghai, China

WORK EXPERIENCE

UC Capital (AUM: US\$500M / NT\$15B)

10/2024 – 12/2025

Machine Learning Quantitative Researcher (full-time until 06/2025, extended as contract role)

Taipei, Taiwan

- Spearheaded the redesign of an institutional trade execution system, ultimately developing a risk-controlled statistical decision framework that improved average execution price by **1 basis point (bp)** ($\approx$ **US\$1.3M** (NT\$40M) annual profit uplift).
- Established the company's first trusted market data foundation by validating and standardizing over **10.8TB** of tick-level trading data (2 years, 1,800 stocks), replacing 12+ fragmented datasets with a centralized source adopted across trading and AI teams.
- Developed an interpretable limit-up stock ranking system by transforming 5 trader-defined heuristic factors and adding 8 features, reducing stock selection time from 2 hours to 30 minutes and improving next-day returns by **0.5%**, yielding **US\$30M+** (NT\$1B) annual trading impact.

Independent

01/2026 – present

Machine Learning Engineer

Taipei, Taiwan

- Quantified the impact of Strait of Hormuz blockade on capital flows and market risk, deriving risk signals for portfolio exposure.
- Developed a reinforcement learning-based thermal control system for laptop OEMs to improve performance stability under dynamic workloads, reformulating a rule-based thermal policy into a sequential decision-making problem to mitigate performance oscillations.

RESEARCH PROJECTS

Diffusion-Reward Adversarial Imitation Learning

04/2023 – 08/2024

Co-first Author and Project Lead, Paper, Project Page, Code

Neural Information Processing Systems (NeurIPS) 2024

- Spearheaded a joint research project with a senior NVIDIA researcher, driving research direction and cross-team execution.
- Identified a flawed modeling assumption through **first-principles analysis**, pivoted the research direction, and transformed a stalled project into a NeurIPS 2025 publication.

Diffusion Model-Augmented Behavioral Cloning

06/2022 – 05/2024

Co-first Author and Project Lead, Paper, Project Page, Code

International Conference on Machine Learning (ICML) 2024

- Initiated and validated a diffusion-based imitation learning framework, leading the full research cycle from idea formulation to empirical validation, including iterative refinement under reviewer feedback.
- Expanded the lab's research directions, inspiring **5+ follow-up projects**, including 2 accepted at top-tier conferences or adopted in industry collaborations, while coordinating research execution across PhD students and interns.

EXTRACURRICULAR PROJECTS & LEADERSHIP

Autonomous Driving Team, Algorithm Group

08/2019 – 12/2020

Algorithm Team Leader, Competition

Shanghai, China

- Spearheaded a LiDAR-SLAM + lightweight YOLO-based perception system for an autonomous racing vehicle, enabling real-time obstacle avoidance under embedded constraints and improving detection success rate from 60% $\rightarrow$ 88%.
- Implemented a closed-loop data iteration framework (logging, relabeling, retraining failures) and coordinated cross-functional hardware integration (sensor placement, latency, compute limits), while supervising junior members during system testing and validation.

RoboMaster Robotics Team, Algorithm Group, Second Place in 2019 National Competition 12/2018 – 09/2019

Algorithm Team Member, Competition, Code

Shanghai, China

- Collaborated on real-time perception and control systems for armor detection and targeting under competition constraints.
- Diagnosed auto-aim instability via camera calibration and perception debugging, separating perception from sensor-related issues.

University Debate Team

09/2018 – 02/2020

Debater and Peer Coach

Shanghai, China

- Mastered debate techniques via Freshman Cup training; selected to represent the department in university-wide competitions.
- Mentored **8+ junior debaters**, designing and delivering **3+ training workshops** on rebuttal strategies and critical reasoning.

SKILLS/QUALIFICATIONS

Programming & Systems

Python (advanced), C/C++ (coursework), Linux, Git, Docker

Machine Learning

Deep Learning (PyTorch), NLP (HuggingFace), Computer Vision (OpenCV), ML (Scikit-learn)

Research Areas

Reinforcement Learning, Imitation Learning, Diffusion Models

Data & Infrastructure

NumPy, Pandas, SQL (SQL Server, MariaDB), Weights & Biases

Interests & Languages

Snowboard (intermediate), LEGO (50+ sets built), Tennis (beginner), English (TOEFL 94)